

MB 554: Blockchain for Business

Course Code	MB 554
Course Name	Blockchain for Business
L-T-P-C	2-0-0-2
Prerequisites	None
Intended for	MBA
Mutual Exclusion	None
Approval	55th BoA (May 2024)

Course Contents

- **Introduction to Blockchain:** Basic ideas behind blockchain, its purpose, how it is changing the landscape of Business, Enterprise Blockchain, Why Blockchain matters? Public vs Private vs Permissioned Blockchains and use cases, Blockchain Questions from Business and Technology Leaders, Blockchain as a Service. (4 Hours)
- **Cryptographic Concepts Required:** Confidentiality, Integrity, Authentication, Mathematics, Symmetric key Cryptography-Stream and block ciphers, Asymmetric key Cryptography, Discrete logarithm, RSA, Hash function, Digital Signature - ECDSA. (4 Hours)
- **Distributed Consensus:** The real need for mining – consensus – Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Byzantine Generals Problem, Byzantine fault tolerance, Sybil Attack, Energy utilization and alternate, Business applications (appropriate case studies, use cases and situation analysis). (4 Hours)
- **Cryptocurrencies – business use cases of blockchain technology:** Introduction to Cryptocurrency, what is a Bitcoin? Double Spending Problem, Bitcoin Mining, Transactions, The Bitcoin network, Bitcoin payments, Bitcoin ICO- Advantages and Disadvantages, Ethereum and its basic Features, The Ethereum network, Components of the Ethereum ecosystem, Ethereum Virtual Machine, Gas, Applications Built based on Ethereum, ETH, Smart Contracts, Smart contract templates, DApp, Business applications (appropriate case studies, use cases and situation analysis). (8 Hours)
- **Introduction to Hyperledger:** Projects under Hyperledger, Permissioned Blockchain and use cases, Hyperledger as a protocol, The reference architecture, Privacy and confidentiality, Scalability, Hyperledger Fabric. (4 Hours)
- **Blockchain Use Cases:** Supply Chain Management, E-Governance, Land Registration, Medical Information Systems, and others. (4 Hours)

Text Books

1. NA

References

1. Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Goldfeder, S., **Bitcoin and cryptocurrency technologies: a comprehensive introduction**, Princeton University Press, 2016.
2. Arun, J. S., Cuomo, J., & Gaur, N., **Blockchain for business**, Addison-Wesley Professional, 2019.
3. Nakamoto, S., **Bitcoin: A peer-to-peer electronic cash system**, Decentralized business review, 2008.
4. Tyagi, S. S., & Bhatia, S. (Eds.), **Blockchain for Business: How it Works and Creates Value**, John Wiley & Sons, 2021.
5. Forouzan, B. A., & Mukhopadhyay, D., **Cryptography and network security** (Vol. 12), Mc Graw Hill Education (India) Private Limited, 2015.
6. Bashir, I. (2017). Mastering blockchain. Packt Publishing Ltd.